

AI Invoice Processing

End-to-end n8n pipeline: Gmail/Drive intake, Gemini extraction, Slack approvals, Google Sheets ledger — technical & process documentation

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This report documents a production-ready n8n automation that turns incoming invoices and receipts — arriving either as Gmail attachments or dropped into a Google Drive folder — into validated, categorised, and approved ledger entries. Gemini reads each document, a validation layer catches extraction errors before they reach the books, and Slack handles both error alerts and interactive approve/reject decisions for larger invoices.

1. Executive Summary

Manually keying in invoices is slow and error-prone, and a purely automated pipeline risks posting bad data straight to the books. This workflow strikes a middle ground: every document is read by AI, checked against basic arithmetic and required fields, deduplicated against the existing ledger, auto-categorised, and then either logged straight through (small, routine invoices) or routed to a human via interactive Slack buttons (anything over the configured threshold) — so the business gets speed on the easy cases and a real approval trail on the ones that matter.

Metric	Detail
Intake channels	Gmail (unread invoice/receipt/bill emails with attachments) and a watched Google Drive folder
Extraction engine	Google Gemini, reading PDFs and images directly and returning structured invoice JSON
Validation	Required fields present, and subtotal + tax must reconcile with the stated total
Duplicate protection	The ledger is searched by invoice number before anything new is logged
Categorisation	Gemini assigns one of six expense categories from the vendor and line items
Approval threshold	Invoices up to \$500 auto-approve; anything higher requires a Slack approve/reject decision
Approval timeout	An unanswered Slack approval resolves to “No Response” after 3 days
Failure alerts	A dedicated error trigger posts to Slack if the workflow itself fails

2. Business Value

- No manual data entry — Gemini reads the PDF or photo directly and returns structured, ledger-ready fields.
- Bad extractions caught before they're trusted — a validation step checks the arithmetic and required fields, flagging anything suspicious to Slack instead of silently logging it.
- No duplicate invoices — every new document is checked against the ledger by invoice number before it's processed further.
- Spend control without bottlenecking small purchases — routine invoices under the threshold post automatically, while larger ones get a real human decision via one-tap Slack buttons.
- A living audit trail — every invoice's status (Approved, Pending Review, Rejected, No Response) is tracked in one ledger, with the original approver's decision captured.

3. Architecture Overview

Two independent triggers feed one shared processing pipeline, plus a dedicated error-handling branch that runs whenever the workflow itself fails.

Branch	Trigger	Purpose
Email intake	Gmail trigger, polling every minute	Picks up unread emails matching invoice/receipt/bill with an attachment
Drive intake	Google Drive trigger, on new file	Picks up any file dropped into a designated invoices folder
Shared pipeline	Fed by either intake branch	Extraction, validation, dedup, categorisation, approval routing and ledger logging
Error handling	n8n Error Trigger, on workflow failure	Posts a Slack alert naming the failed workflow and execution

4. Step-by-Step Process

1 Watch two intake channels

A Gmail trigger polls for unread emails matching “invoice”, “receipt” or “bill” with an attachment, while a separate Google Drive trigger watches a designated folder for new files — either channel can start the same pipeline.

2 Filter to PDFs and images only

Each incoming file's MIME type is checked; anything that isn't a PDF or an image is skipped and flagged to Slack for manual handling rather than sent to the AI blind.

3 Extract with Gemini

Gemini reads the document directly (no separate OCR step) and returns structured JSON: vendor, invoice number, dates, line items, subtotal, tax, total and currency. The call retries automatically on transient failures.

4 Parse & validate the extraction

The AI's JSON is parsed defensively, required fields (vendor, invoice number, a positive total) are checked, and the subtotal plus tax is verified to reconcile with the stated total within a small tolerance. Anything that fails is flagged as needing review and posted to Slack with the reason, rather than logged as-is.

5 Check for duplicates

The ledger is searched by invoice number; a match skips the invoice with a Slack notification instead of creating a duplicate entry.

6 Classify the expense category

A second, lightweight Gemini call assigns exactly one category — Software, Utilities, Contractor, Office Supplies, Travel or Other — based on the vendor and line items.

7 Route by approval threshold

Invoices at or under the configured limit (\$500 by default) are marked Approved and logged straight to the ledger. Anything above it is marked Pending Review and logged before any approval request goes out, so it's tracked even while awaiting a decision.

8 Request approval in Slack

For invoices needing review, an interactive Slack message with Approve / Reject buttons is posted, showing the vendor, invoice number, total, category and due date; if no one responds within 3 days the request times out automatically.

9 Record the decision

The clicked outcome (or the 3-day timeout) is translated into Approved, Rejected or No Response, and the ledger row is updated by invoice number to reflect the final status.

5. Node-by-Node Map

Stage	Key nodes
Email intake	Watch Invoice Emails → Email Attachment Is PDF or Image? → Extract Invoice Data (Email) → Tag Source Email → Mark Email As Read
Drive intake	Watch Invoice Drive Folder → Download Drive File → Drive File Is PDF or Image? → Extract Invoice Data (Drive) → Tag Source Drive
Merge & validate	Combine Extraction Results → Parse & Validate Extraction → Extraction Valid? → Notify Extraction Needs Review
Dedup & categorise	Check Duplicate Invoice → Duplicate Found? → Notify Duplicate Invoice / Classify Expense Category
Approval routing	Determine Approval Route → Needs Approval? → Mark Auto Approved / Mark Pending Review → Log Invoice To Ledger / Log Pending Invoice
Slack approval	Request Manual Approval → Set Approval Decision → Update Ledger Status
Error handling	On Workflow Error → Alert Workflow Failure

6. Reliability & Security Design

- **Extraction is never trusted blindly** — required fields and basic arithmetic (subtotal + tax = total) are checked before an invoice is treated as valid, with failures routed to a human via Slack.
- **Duplicate-safe by design** — every invoice is looked up by invoice number against the ledger before it's processed further, preventing the same bill from being paid or logged twice.
- **Spend-gated approvals** — the \$500 auto-approve threshold is a single configurable value, and anything above it requires an explicit human decision captured in Slack.
- **Nothing waits forever** — a pending approval automatically resolves to “No Response” after 3 days rather than sitting unresolved indefinitely.
- **Workflow-level failure alerting** — a dedicated Error Trigger ensures the team is notified in Slack if the automation itself breaks, not just if an individual invoice fails validation.

7. Setup Checklist

- Connect credentials for Gmail, Google Drive, Google Sheets, Google Gemini and Slack.
- Create a Google Sheet with an invoices tab and these headers: vendor_name, invoice_number, invoice_date, due_date, subtotal, tax, total, currency, category, source, needsApproval, status — then select it in all four Google Sheets nodes.
- Choose your Drive folder in Watch Invoice Drive Folder and your Slack channel in every Slack node.
- For in-Slack approvals: create a Slack app with the chat:write scope, enable Interactivity with its Request URL set to <https://webhook-waiting-slack>, add the bot token and Signing Secret to the Slack credential, and invite the bot to your channel.
- In workflow Settings, set Error Workflow to this workflow so On Workflow Error actually fires.
- Activate the workflow, then adjust the \$500 threshold in Determine Approval Route and the categories in Classify Expense Category to match your business.

This workflow was designed and built end-to-end in n8n as part of an AI automation portfolio, demonstrating multi-channel document intake, AI extraction with validation, duplicate protection, and human-in-the-loop spend approval.